Insight

TECHNICAL SESSION

REST vs GraphQL vs gRPC

Choosing the right API protocol for the operation in front of you.

WHAT IT IS

An intermediate tour of REST, GraphQL and gRPC through one working demo app: a local or hosted model routes each question to the protocol that fits, and every request is real traffic you can inspect.

dev2next 2026 • Insight

Three protocols. One question. The right call.

01 · THE PROTOCOLS

REST, GraphQL and gRPC compared on real traffic.

02 · THE DEMO

One app, three protocols, every request inspectable.

03 · THE FIELD

Client stories where the choice mattered.

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YOUR PRESENTERS

Introductions

Two Insight architects who design and build these integrations with clients every day.



Chetan Galgali

SENIOR ENTERPRISE ARCHITECT

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SENIOR SOLUTIONS PRINCIPAL

Insight

Today: REST, GraphQL and gRPC on one running application, and the client work where choosing between them mattered.

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WHO WE ARE

Our Reach as a Solutions Integrator

Insight Enterprises helps organizations plan, build, secure and run the technology their business depends on.

FORTUNE 500 (NO. 427) · FOUNDED IN 1988 · HEADQUARTERED IN CHANDLER, ARIZONA

NET SALES

$8.2B

in 2025

TEAMMATES

14,400+

worldwide

CERTIFIED EXPERTS

6,400+

consulting and service delivery

PARTNERS

6,000+

hardware, software and cloud

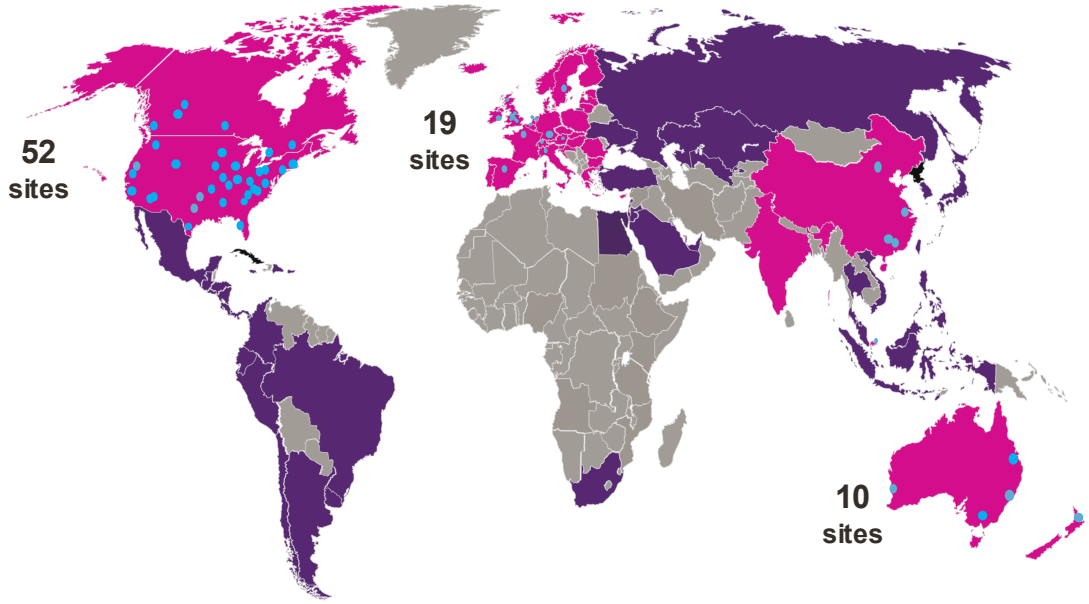
01 Solutions integration

Strategy, design, build and run across cloud, data, AI and security.

02 Technology and supply chain

Source, configure and deploy hardware, software and cloud at scale.

GLOBAL SOLUTIONS DELIVERY



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OUR EXPERTISE

End-to-End Insight Services Across Every Cloud

Six areas of expertise across AWS, Microsoft Azure and Google Cloud, delivered through four service lines.



01 Digital Workplace

Collaboration and productivity platforms, endpoint management, identity, virtual desktops and AI assistants

02 Intelligent Edge

IoT platforms, computer vision, edge computing and connected devices, secured from device to cloud

03 Data and AI

Data platforms and lakehouses, analytics and BI, machine learning, generative AI and data governance

04 Cybersecurity

Security assessments, identity and zero trust, threat detection and response, cloud and network security

05 Cloud Infrastructure

Migration and modernization on AWS, Azure and Google Cloud, hybrid cloud, Kubernetes and cost optimization

06 Apps and Integration

Cloud-native development and modernization, APIs and integration, DevOps, platform engineering and AI-assisted development

Services

Consulting Services

Align to business objectives and leverage technology partner relationships.

Change and Training

Drive adoption of new technologies and optimize business processes.

Software and Lifecycle

Software licensing, cloud marketplace and subscription management, and software asset management.

Managed Services

Managed cloud, apps, data and AI, and security, without disrupting the business.

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TODAY

Key Expectations & Takeaways

Three sections, one running application. Every example is real HTTP, GraphQL and gRPC traffic, with no mocks.

01	**The Landscape**	Why protocol choice is still contentious, the key comparisons, and the demo data set.	*Context*
02	**The Protocols**	REST, GraphQL and gRPC, each covered twice: first the concept, then the working code.	*Two slides per protocol*
03	**Demo & Field Stories**	The live demo, then client stories where the choice mattered.	*Success from the field*
A	**Appendix: AI Routing**	How the demo lets Ollama or Jev choose the protocol, their trade-offs and the guardrails.	*Time permitting*

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01

The Landscape

Why protocol choice is still contentious, and how the three compare.

IN THIS SECTION

Defining the Frontier: the problem every integration team hits

Origins: where each protocol came from and where it fits

Key Comparisons: transport, contract, caching and streaming

Demo Data Set: one catalog behind all three protocols

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WHY THIS MATTERS

Defining the Frontier for Integration

There are many ways to move data between clients, services and partners. Choosing one is rarely obvious.

01 · THE PROBLEM

Every team has a default

Frontend, backend and integration teams each reach for what they know, so the data contract becomes a negotiation.

02 · THE REALITY

Each protocol has a sweet spot

Cacheable resources, client-shaped graphs and low-latency streams are different problems with different best answers.

03 · THE APPROACH

Choose per operation

Most systems benefit from more than one protocol, if each contract stays typed, versioned and visible.

The question is not which protocol wins. It is which one fits the operation in front of you. Most production systems run more than one.

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USAGE COMPARISON

Where Each Protocol Came From, and Where It Fits

Three decades of API design. Each protocol answered the problem of its moment.

2000

REST described in Roy Fielding's dissertation

2012

GraphQL built inside Facebook for mobile

2015

GraphQL spec published; gRPC open-sourced by Google

2016

gRPC 1.0 released; adopted by CNCF in 2017

	REST	GraphQL	gRPC
Learning curve	Low	Medium	Medium: Protobuf and toolchain
Ecosystem	Abundant	Mature and growing	Mature in cloud-native
Typical use cases	Public APIs, identity, legacy integration	Mobile, BFFs, micro-frontends, multi-tenant	Microservices, IoT, real-time streams

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SIDE BY SIDE

Key Comparisons

The same questions asked of each protocol. There are no bad answers, only different fits.

Feature	REST	GraphQL	gRPC
Transport	HTTP/1.1, HTTP/2 or HTTP/3	HTTP, usually one POST endpoint	HTTP/2 (required)
Payload	JSON (any media type)	JSON	Protobuf, binary
Contract	OpenAPI (optional)	SDL schema (required)	.proto file (required)
Data shape	Fixed per endpoint	Client selects fields	Fixed per RPC, strongly typed
Caching	Native HTTP and CDN caching	Client-side or persisted queries	Application-level
Streaming	SSE or chunked responses	Subscriptions	Server, client and bidirectional
Browser support	Native	Native	Needs gRPC-Web or a bridge
Best fit	Public and resource APIs	Client-driven nested data	Service-to-service, low latency

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DEMO DATA SET

One Catalog Behind All Three Protocols

Ship and film data, enriched with selected character appearances and verified flight crew.

STARSHIPS

19

Name, max speed and length, from the CR90 corvette to the Millennium Falcon.

many-to-many



FILMS

6

A New Hope through Revenge of the Sith.

many-to-many



CHARACTERS

18

Luke Skywalker, Han Solo, Leia Organa and Chewbacca. A curated selection.

One immutable, in-memory catalog. REST, GraphQL and gRPC all read the same seeded source. There is no database to configure, and all three protocols see identical data.

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02

The Protocols

REST, GraphQL and gRPC: each in concept, then in working code.

IN THIS SECTION

REST: resources over HTTP

GraphQL: client-shaped queries over one schema

gRPC: typed contracts and streaming over HTTP/2

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REST · HOW IT WORKS

REST: Resources Over HTTP

Every resource gets a URL and HTTP verbs carry the intent. Its simplicity is both the strength and the ceiling.

Where REST shines

- Addressable resources with standard verbs and status codes
- Native HTTP caching through browsers, CDNs and proxies
- Universal tooling: curl, OpenAPI, gateways and API management
- The lowest barrier for partners and public consumers

Where it strains

- Over-fetching: fixed payloads return fields you do not need
- Under-fetching: related data means extra round trips
- Versioning pressure as client needs diverge
- No first-class streaming (SSE or polling instead)

Use REST for simple, cacheable, stateless resource APIs, especially ones you publish to others.

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REST · IN THE DEMO

REST in the Demo

Single-resource lookups route here: one URL per resource and a cacheable JSON response.

REQUEST · RESPONSE

```
GET /api/catalog/starships/ship:5

200 OK
Cache-Control: max-age=300
[
  {
    "id": "5",
    "name": "Millennium Falcon",
    "maxSpeed": 1050,
    "length": 34.37,
    "films": [ … ],
    "crew": [ … ]
  }
]
```

01 One route per resource

/starships/{target}, /films/{id} and /cast/{id} on Spring MVC.

02 Cacheable by default

Starship reads send Cache-Control: max-age=300.

03 Picked for simple lookups

The router sends speed, length and detail questions here.

Try it: “Tell me about the Millennium Falcon.”

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GRAPHQL · HOW IT WORKS

GraphQL: Ask for Exactly What You Need

One endpoint and one typed schema. The client decides the shape of the response.

Where GraphQL shines

- A single typed schema (SDL) that describes the whole graph
- Clients select exactly the fields a screen needs
- Nested relationships resolved in one round trip
- Introspection powers tooling, docs and code generation

Where it strains

- HTTP caching is harder: POST to one URL
- N+1 resolver load without batching such as DataLoader
- Public APIs need query depth and cost limits
- More server-side complexity to operate and secure

Use GraphQL for client-driven screens that stitch related data: BFFs, mobile apps and micro-frontends.

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GRAPHQL · IN THE DEMO

GraphQL in the Demo

One query returns a film's ships and the crew who flew them. Cast questions use a separate operation.

POST /GRAPHQL

```
query FilmShipsCrew($target: String!) {
  films(target: $target) { id title }
  starships(target: $target) {
    name
    crew { filmId role member { name } }
  }
}

# variables: { "target": "film:1" }
```

01 **One request, linked records**

Film title, ship names and crew roles in one query.

02 **Measured against REST**

Both need one request. GraphQL selects fields, while REST includes nested records.

03 **Crew, not cast**

Crew comes from documented assignments. Appearing in a film does not mean piloting.

Try it: “Which starships appear in A New Hope, and who flew them?”

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GRPC · HOW IT WORKS

gRPC: Typed Contracts, Streaming by Default

Contract-first RPC over HTTP/2 with compact Protobuf messages and four call types.

Where gRPC shines

- Contract-first .proto files with generated clients in many languages
- Compact binary Protobuf with HTTP/2 multiplexing
- Unary, server, client and bidirectional streaming
- Deadlines, cancellation and metadata built in

Where it strains

- Browsers need gRPC-Web or an HTTP bridge
- Binary payloads are harder to read and debug
- HTTP/2 end to end through proxies and load balancers
- Schema discipline: field numbers are forever

Use gRPC for internal service-to-service calls, low-latency paths and real-time streams.

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GRPC · IN THE DEMO

gRPC in the Demo

Track a simulated Falcon flight: one request, 21 updates, about 40 seconds.

CATALOG.PROTO

```
service StarshipTracker {
  rpc Watch(WatchRequest)
      returns (stream TrackSnapshot);
}
message WatchRequest {
  string target = 1;
}
message TrackSnapshot {
  repeated TrackedShip starships = 1;
  int32 sequence = 2;
  string observed_at = 3;
  FlightState flight = 6;
}
```

01 Subscribe once

One request starts a server stream; location, pilot and shields change.

02 Inspectable updates

The timeline pairs each Protobuf payload, byte count and arrival time by sequence.

03 Bounded and honest

A labeled, repeatable simulation. Stop anytime; replay the same flight.

Try it: “Track the Millennium Falcon.”

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03

Demo & Field Stories

The live demo, then where the choice mattered for clients.

IN THIS SECTION

Live demo: three questions, three protocols

Port terminal operator: REST and GraphQL

Global retail coffee: gRPC at the edge

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LIVE DEMO

Demo: Let's Get Into the Code

Ask a question, see the protocol and why it fits, then inspect the real request and response.

REST · SIMPLE LOOKUP

“Tell me about the Millennium Falcon.”

Routes to GET /api/catalog/starships/ship:5 and returns a cacheable JSON body.

GRAPHQL · RELATIONSHIPS

“Which starships appear in A New Hope, and who flew them?”

FilmShipsCrew returns the film’s ships and their documented crew.

GRPC · STREAMING

“Track the Millennium Falcon.”

One Watch subscription streams a 40-second simulated flight into an updating chat card.

RUN IT

```
docker compose up --build -d
open http://localhost:3000
```

BUILT WITH

React 19 · TypeScript · assistant-ui · Zustand · Tailwind v4
Java 25 · Spring Boot 4.1 · Spring for GraphQL · Spring for gRPC
Ollama (local) · Jev (optional, hosted)

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CLIENT STORY · LOGISTICS

Port Terminal Operator

A leading U.S. marine terminal operator: a corporate website its marketing team can run.

The client needed a modern, fast website that marketing could update without waiting on IT.

The approach

- UX workshops with key stakeholders to map user journeys and critical paths
- Wireframes refined side by side with stakeholders
- Modern web front end on managed cloud hosting, with zero-downtime deployments
- SaaS headless CMS with reusable components, so marketing builds its own pages

Why it matters here: A headless CMS is an API product. Choosing REST or GraphQL per page is the same decision the demo makes per question.

PROTOCOLS IN PLAY

REST and GraphQL

The front end consumes CMS content through REST and GraphQL APIs, decoupling content from presentation.

RESULTS

Days to hours

New pages now take hours instead of days. Image payloads are 45% smaller with no visible quality loss, and organic search traffic improved.

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CLIENT STORY · RETAIL

Global Retail Coffee Manufacturer

Next-generation point of sale and order operations.

The client set out to build the next-generation POS and ordering system.

LOCATIONS

33k

security and offline operation are key

WHERE IT IS GOING

The drive-thru

Now engaged on opportunities to improve drive-thru experiences.

Why gRPC

- Better experiences for both the barista partner and the customer
- Limited ability to expose endpoints at the store
- gRPC integration reduces exposure and increases performance
- Service-based integration built on typed gRPC contracts

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WHAT TO TAKE AWAY

Three Things to Take Away

Pick the protocol per operation, design the contract first, and measure the whole path.

01

Choose per operation

REST for cacheable resources, GraphQL for related data, and gRPC for streams and service-to-service calls. Most systems use more than one.

02

Design the contract first

OpenAPI, SDL or .proto: the contract outlives any implementation and drives tooling, testing and code generation.

03

Measure the whole path

Round trips, payload size and caching decide real-world latency more than the protocol label does.

“Three protocols. One question. The right call, made where you can see it.”

Repository, runbook and measurements: https://github.com/spietrek-insight/rest-graphql-grpc-ai-routing

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THANK YOU

/questions

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Steve Pietrek | Senior Solutions Principal

Try it yourself.

01 · RUN

docker compose up --build -d, then open localhost:3000.

02 · ASK

Use the three guided prompts, then ask your own.

03 · INSPECT

Open the inspector to see every payload and timing.

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A

Appendix: AI Routing

Reference material: how the demo lets a model choose the protocol.

IN THIS SECTION

Architecture: how a question becomes a protocol call

Ollama: local tool calling on a 4B model

Jev: structured decisions with confidence

Trade-offs and guardrails

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ARCHITECTURE

How the Demo Routes a Question

One interface, IntentClassifier, with two implementations: Ollama by default and Jev optionally.

01 · ASK

A plain question

The browser posts the message and recent turns to /api/chat.

02 · CLASSIFY

Ollama or Jev

Picks one protocol and one allowlisted catalog target.

03 · DECIDE

A typed decision

Only { protocol, target }: never a URL or query text.

04 · EXECUTE

Real transport

Calls its own REST, GraphQL or native gRPC endpoint.

05 · ANSWER

Compose, inspect

Built from the response, with payloads and timings shown.

The model chooses the route. It never supplies the data. Answers are composed deterministically from what the protocol returned, so a routing mistake is visible and no facts are invented.

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AI ROUTING · OLLAMA

Local Routing with Ollama Tool Calling

Genuine tool calling on a small local model: no API key and no cloud dependency.

ONE TOOL PER OPERATION

```
{
  "type": "function",
  "function": {
    "name": "FILM_SHIPS_CAST",
    "description": "Ships and film cast",
    "parameters": {
      "properties": { "target": {
        "type": "string",
        "enum": ["film:1", …, "none"]
      } },
      "required": ["target"]
} } }
```

01 The tool chooses the operation

Each reviewed operation fixes its protocol and request.

02 Enum-bound targets

Character IDs and ship/film crew targets stay allowlisted.

03 A pinned small model

qwen3:4b-instruct-2507 (q4_K_M), no thinking mode, runs locally.

04 Tuned for latency

Temperature 0, 96-token cap, 30-minute keep-alive, startup warm-up.

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AI ROUTING · JEV

Jev: A Decision Model, Not a Text Generator

Two typed Choice questions replace free-form generation, which suits a small, closed decision space.

POST /V1/SYSTEMONE

```
{
  "model": "jev-latest",
  "state": "Context … Latest question …",
  "questions": {
    "operation": { "type": "choice",
      "criteria": {
        "FILM_SHIPS_CAST": "…",
        "SHIP_CREW": "…", "TRACK": "…" } },
    "target": { "type": "choice",
      "criteria": { "film:1": "…" } }
  }
}
```

01 Answers, not prose

Each question returns a choice with confidence and probabilities, shown in the inspector.

02 Same contract as Ollama

Plugs in behind IntentClassifier with AI_PROVIDER=jev.

03 Hosted by TypeSafe

Needs internet and TYPESAFE_API_KEY, and the key never leaves the server.

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CHOOSING A ROUTER

Ollama vs Jev

Two ways to make the same decision. Swap between them with one environment variable.

	Ollama (default)	Jev (optional)
Mechanism	Tool calling across reviewed operations	Two typed Choice questions
Where it runs	Locally, in Docker or native on the host	Hosted TypeSafe API
Output	One tool call plus an enum target	A choice per question
Confidence	None reported, and none invented	Confidence and probabilities
Setup	Pull a model of several GB once	API key and internet access
Best for	Offline demos and data residency	Closed decisions that need calibrated confidence

Presenter overrides bypass the model entirely for one question, and the UI says so. They are never passed off as AI classification.

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TRADE-OFFS

Ollama vs Jev: Pros and Cons

Same decision, different trade-offs. Pick the router that fits your constraint.

	Ollama (default)	Jev (optional)
Pros	• No API key, no cloud dependency, works offline • Questions never leave the machine • Genuine tool calling with enum-bound targets • A pinned model gives repeatable routing	• Built for closed decisions: typed choices, not free text • Confidence and probabilities on every decision • No model download or local GPU needed • Same interface: one environment variable to switch
Cons	• Multi-gigabyte model download, and a slow first question while it loads • Routing speed depends on local hardware • No confidence score to judge a close call • A small model can misroute an ambiguous question	• Needs internet access and an API key • The question and recent turns go to a hosted third-party service • Latency and availability depend on that service • No offline mode for demos or restricted networks

Pick by constraint Offline demos and data residency point to Ollama. Calibrated confidence points to Jev. Either way, a failure shows an error, never a silent fallback.

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GUARDRAILS

Five Rules That Keep the Router Honest

The model gets a narrow job, and the system makes every choice it makes visible.

01	**Allowlisted targets**	Models choose catalog IDs through an enum or criteria, never URLs, queries or service names.	*Blocks injected or hallucinated calls*
02	**No hidden fallback**	Invalid output or an unavailable provider shows an error. Nothing is keyword-routed and labeled as AI.	*What you see is what the model did*
03	**Labeled overrides**	Presenter overrides apply to one question, then the router resets to Auto.	*Demos never stall on a cold model*
04	**Grounded answers**	Every answer is composed from the actual transport response.	*No invented facts*
05	**Secrets stay server-side**	Keys and auth headers never reach the browser. Ports bind to loopback.	*Safe local-first defaults*

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